

Competition and Permanent Reputations*

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Abstract

In the canonical dynamic reputation game with a single long-lived player, Cripps, Mailath, and Samuelson (2004) show that reputations for non-credible behavior eventually vanish. We revisit this question in a setting with two long-lived players competing to interact with a sequence of short-lived players. We show that when monitoring is sufficiently informative, competition for reputation sustains permanent reputation maintenance in all pure strategy Markov equilibria. The key force is that payoffs depend on the ranking of reputations rather than their absolute levels, and this gives each player a persistent incentive to maintain an advantage over his opponent.

1 Introduction

Whether an agent is chosen to perform a task is typically tied to how reliable he is perceived to be. Consumers buy products from firms that are trusted to have high quality standards, homeowners hire handymen they believe will work hard, managers assign tasks to more reliable agents. Being perceived as reliable often requires exertion of some costly effort. However, if the agent cares enough about his future earnings, he would be willing to pay this cost.

The incentives created by reputational concerns have been studied extensively since the seminal contributions of Milgrom and Roberts (1982), Kreps and Wilson (1982), and Fudenberg and Levine (1989). In the canonical model, a single long-lived player repeatedly interacts with a sequence of short-lived players. The long-lived player may be either a strategic type, who can freely choose his actions, or a commitment type, who mechanically follows a fixed course of behavior. After each interaction, a public signal provides imperfect information about the player's type. Because future opponents condition their behavior on this information, a strategic player may find it profitable to mimic the commitment type and build a reputation that can later be exploited.

However, Cripps, Mailath, and Samuelson (2004) show that these reputational incentives eventually disappear. They consider a setting in which the commitment type repeatedly plays the same action.

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Their main result is that if this action is not part of a stage-game Nash equilibrium, then the strategic player's type is revealed in every equilibrium of the repeated game. The intuition is that the short-lived player's belief is a martingale and therefore converges almost surely. As a result, new information eventually has an arbitrarily small effect on the strategic player's reputation and future payoffs. Once reputational concerns become sufficiently weak, the short-run gain from deviating outweighs the future cost of doing so, causing the strategic player to abandon the commitment behavior.

This paper shows that competition between two long-lived players can prevent reputations from vanishing. We study a model with a sequence of short-lived consumers (she) and two long-lived agents (he). Each consumer hires one of the agents. Agents receive a positive payoff when hired. Once an agent is selected, he chooses whether to shirk or exert effort at a cost. This choice is unobservable to the consumer. In a one-shot interaction, the agent strictly prefers shirking conditional on being hired. Therefore there is no Nash equilibrium of the stage game in which the chosen agent exerts effort.

Each agent has a privately known type. He is either reliable, meaning he commits to exerting effort if hired, or he is strategic, and can choose either to work or shirk. The hired agent's effort determines the distribution of output that the consumer receives. High effort yields higher expected output. Since she is short-lived, the consumer hires the agent she believes is most likely to exert effort. This is jointly determined by the agent's reputation as well as her expectations about the behavior of strategic types.

If there were only one agent, then, as Cripps, Mailath, and Samuelson (2004) show, reputational incentives would eventually dissipate. As beliefs converge, any single output realization has an increasingly small effect on future reputations and payoffs. With multiple agents, however, even small changes in reputation can affect which agent is perceived as more reliable. Consequently, each agent continues to value marginal improvements in reputation because these improvements help maintain an advantage over his rival. We establish that when effort costs are sufficiently low and feedback is sufficiently informative, the benefits of having the highest reputation are large enough to prevent reputations from vanishing in any perfect Bayesian equilibrium with pure strategy Markov agent strategies.

We focus on pure strategy Markov agent strategies because under this restriction, equilibria admit a straightforward structure. As shown in section 4, all pure strategy Markov equilibria start with a *shirking* phase in which strategic agents shirk when hired, and consumers select the agent with the higher reputation. The game then transitions to a *working* phase, where both agents exert effort independent of type. Once the game enters the working phase, reputations remain static and hence do not vanish.¹ If the working phase is never reached, then reputations of both agents almost

¹This is a consequence of imperfect monitoring. Reputations must be strictly higher than 0 after any finite number of interactions.

surely vanish.² The main result of the paper offers a sufficient condition for the working phase to be almost surely reached in finite time.

The argument is via a contradiction. Suppose the working phase is not reached in finite time: strategic agents shirk forever. Then the agent with the higher reputation today will continue to be hired until his reputation falls below his opponent's. Since he is constantly shirking, there will almost surely arrive a pivotal history: one in which the agent has a higher reputation today, but he will not if the consumer receives one more bad output level. By deviating to high effort, the agent with the higher reputation improves the likelihood that output is good, and thus the likelihood of maintaining his advantage. Assuming this probability difference is large enough relative to the cost of effort, this deviation is profitable for sufficiently patient agents. As a result, even if reputations become arbitrarily small, incentives to preserve a reputational advantage ensure that the working phase is reached in finite time, thus preventing reputations from going to 0.

We pause to discuss some limitations of the pure strategy Markov restriction. The proof of the main result exploits the shirk-then-work structure of equilibrium. When agents are allowed to mix, for instance, there are equilibria in which reputations vanish. However, such equilibria rely on indifference conditions that neutralize the incentives generated by reputation rankings. Ultimately, though, we do not have conclusive arguments for properties of equilibria outside the pure Markov class. This is discussed in further detail in section 3.

Any strategy profile in a subgame starting in the working phase must itself be sustained as a perfect Bayesian equilibrium. We therefore characterize when a first-best equilibrium—one in which both agents exert high effort in every period—exists. We provide a necessary and sufficient condition for existence, together with a simple consumer strategy that supports such an equilibrium. This establishes the existence of an equilibrium in which reputations never vanish. The intuition is straightforward: when both types exert effort, output is uninformative and reputations remain constant. The consumer sustains effort by conditioning future hiring decisions on output rather than reputation. Unlike our main theorem, this result does not rely on the pure strategy Markov restriction.

In section 2, we discuss related literature. In section 3, we present the model. In section 4, we characterize the structure of all equilibria. In section 5, we show when there exists a first-best equilibrium. In section 6, we discuss equilibrium asymptotics, where we present our main result. Section 7 discusses an extension, and we conclude in section 8.

2 Related Literature

This paper contributes to the reputations literature initiated by Kreps and Wilson (1982), Fudenberg and Levine (1989), Fudenberg and Levine (1992) and others. Fudenberg and Levine (1989) study ex-ante payoffs in a repeated game between a long-lived player and a short-lived player. Our

²This follows from the logic of Cripps, Mailath, and Samuelson (2004).

focus is on asymptotic behavior. In this respect, our main result relates most closely to that of Cripps, Mailath, and Samuelson (2004). In particular, we show that under competition, even small changes in reputation can have large payoff consequences because they may alter which agent is perceived as more reliable. Consequently, reputations do not vanish in equilibrium.

Several papers identify environments in which reputations do not vanish (Liu (2011), Board and Meyer-ter-Vehn (2013), Wiseman (2008), Phelan (2006)). In Liu (2011), this is driven by limited memory of the short-run players. In Board and Meyer-ter-Vehn (2013), it is a consequence of a dynamically changing environment. In contrast, we focus purely on competition among long-lived players.

Two closely related papers that discuss competition and reputation are Hörner (2002) and Deb and Fanning (2026). Hörner (2002) studies a competitive market with good firms and bad firms. The main result studies when there exists an equilibrium in which strategic firms always exert high effort on path. This is similar to our first-best existence result, but we additionally prove that it is impossible for agents to shirk indefinitely in any pure strategy Markov equilibrium. Deb and Fanning (2026) study a similar setup with finitely many firms with the additional feature that firms that do not receive business in any given period are driven out of the market. In their model, persistent high effort is impossible in equilibrium because eventually a monopolist emerges. In our model, the emergence of a monopolist is impossible. Therefore, it is possible to incentivize high effort indefinitely because the consumer can always threaten an agent to take her business elsewhere.

Kartik, Lipnowski, and Pei (2025) also study whether replacement can discipline agent behavior. In their model, there is an infinite pool of agents, and once the consumer chooses to fire an agent, he may never be rehired. Their main finding is that there exists a first-best equilibrium if and only if there exists an equilibrium that does not eventually attain first-best. In our model, this relationship does not hold. We identify conditions under which first-best existence implies eventual first-best in all pure strategy Markov equilibria, and conditions under which the converse implication holds. One reason our results differ is that we focus on pure Markov strategies whereas their results speak to a larger class of equilibria. Furthermore, in their model, eventual first-best is attained because the consumer eventually settles on a reliable agent. In our model, the existence of such an agent is not guaranteed. Therefore, the only way for the short-lived player to achieve eventual first-best is for the strategic agent to eventually find it incentive compatible to exert high effort.

Our paper also relates to the literature on dynamic agent selection and delegation (De Clippel et al. (2021), Andrews and Barron (2016), Board (2011)). Each of the papers studies how a principal exploits competition between agents to incentivize her most-preferred outcome in equilibrium. Lipnowski and Ramos (2020) also study a dynamic delegation problem, but they study a consumer who chooses between hiring an agent and an outside option in every period. In our setting, the consumer uses an agent's opponent as an outside option to incentivize high effort.

From a technical standpoint, our results rely on the martingale convergence theorem, dynamic programming techniques from Stokey, Lucas Jr, and Prescott (1989), and recursive methods from Abreu, Pearce, and Stacchetti (1990).

3 Model

We consider an infinite horizon game played between a sequence of short-lived³ consumers and two long-lived agents with common discount factor $\delta < 1$. Time is discrete and indexed by $t = 0, 1, 2, \dots$. The structure of the stage game is as follows:

Stage Game: At the beginning of each period t , the consumer chooses an agent to hire. Denote the consumer's choice of agent in period t by x_t . The chosen agent receives a payoff of 1 whereas the other receives a payoff of 0. If selected, agent i then chooses an effort level $e_{it} \in E := \{0, 1\}$ at a cost ce_{it} . His effort choice is unobservable to the consumer. However, at the end of period t , the consumer receives a publicly-observed output $y_t \in [0, \bar{y}]$ where $\bar{y} > 0$.

Output: The distribution of output depends on the effort chosen. Let F_e denote the distribution of output under effort level e . Assume F_e is continuous and has a strictly positive and continuous density f_e . Implicitly, this means there are no output levels that perfectly reveal the agent's choice of effort. We assume that output satisfies the strict monotone likelihood ratio property. Formally,

Assumption 1. $y' > y \Rightarrow \frac{f_1(y')}{f_0(y')} > \frac{f_1(y)}{f_0(y)}$

By Assumption 1, F_1 first-order stochastically dominates F_0 . We normalize the expected output under high effort to 1 and denote the expected output under low effort by $\underline{y}$. Furthermore, Assumption 1 implies the existence of unique y_* such that $f_1(y) \geq f_0(y)$ if and only if $y \geq y_*$.

Payoffs and information: Each agent has a privately-known type. With probability λ_i^0 , agent i is *reliable*, which means that he will choose $e_{it} = 1$ in each period t independent of the history. With probability $1 - \lambda_i^0$, agent i is *strategic* and can choose either effort level in each period. The strategic agent i chooses actions to maximize:

$$(1 - \delta) \sum_{t \geq 0} \delta^t \mathbb{1}\{x_t = i\} (1 - ce_{it}).$$

At the beginning of period t , the consumer observes an element of the set $H_t := (\{1, 2\} \times Y)^t$, which specifies the sequence of agents hired in previous periods as well as their output levels.⁴ The consumer updates her beliefs about agent types via Bayes' rule. By independence, output generated by agent i offers no information about his opponent's type.

³Consumers being short-lived is common in the reputations literature. There is work that features only long-lived players, but given that our central point of comparison is Cripps, Mailath, and Samuelson (2004), we adhere to their modeling choices with the exception of there only being one long-lived agent.

⁴Information sets for all players are formally defined in the appendix.

The consumer's information is public. In addition to the public history, each agent observes his own private action in each period he was hired. Agent i does not observe his opponent's effort choices.

Strategies and beliefs: Let λ_i^t be the consumer's period t belief that agent i is reliable. Throughout the rest of the paper, we refer to this as agent i 's period t reputation.

The consumer's strategy is a map from each public history to a choice of agent. Because the consumer is myopic, in each period, she derives a higher expected payoff by hiring the agent who offers the highest probability of exerting high effort. This is jointly determined by two things: each agent's reputation as well as the behavior of strategic agents, which we define below.

Agent strategies are Markov: agent i 's strategy is a function $\sigma_i : [0, 1]^2 \rightarrow \{0, 1\}$ that maps current agent reputations to an effort level. We interpret σ_i as agent i 's effort choice for any pair of reputations given that he is strategic.

Equilibrium: We study Perfect Bayesian Equilibria of this game with pure Markov agent strategies. The formal definition can be found in the appendix. In the following sections, we refer to all pure Markov PBE simply as equilibria.

3.1 Comments on the Model

Prior to a discussion of the results, we make a few comments.

1. **Independence:** We have assumed agent types are independently distributed. Theorem 1 is not contingent on type correlation, but Theorem 2 is. We show in section 7 that Theorem 2 can fail if agent types are positively correlated but is sustained if agent types are negatively correlated.
2. **Pure strategies:** We are interested in whether having a higher reputation than one's opponent induces high effort in the long run. In pure strategy equilibria, when both strategic agents shirk, hiring decisions are determined entirely by reputation rankings. Allowing for mixed strategies introduces an additional margin: the probability of effort conditional on being strategic. Therefore, there may exist equilibria in which lower reputation agents offer a higher total probability of high effort. Such equilibria may possess interesting properties, but they are beyond the scope of this paper.

4 Equilibrium Analysis

In this section, we show that pure strategy Markov equilibria admit a simple structure: the game starts in a shirking phase, where each agent shirks conditional on being strategic. The game then transitions to a working phase, where each agent works in every period irrespective of type. We begin by characterizing how reputations evolve over time. Consider an equilibrium strategy profile σ .

Suppose in period t , agent 1 is hired and y_t is realized. In period $t + 1$, his reputation is updated via Bayes' Rule:

$$\lambda_1^{t+1}(y_t) = \frac{\lambda_1^t f_1(y_t)}{\lambda_1^t f_1(y_t) + (1 - \lambda_1^t)(\sigma_1(\lambda_1^t, \lambda_2^t) f_1(y_t) + (1 - \sigma_1(\lambda_1^t, \lambda_2^t)) f_0(y_t))}. \quad (1)$$

Since types are independently distributed, if agent 1 is hired in period t , output offers no information on agent 2's type. In other words, $\lambda_2^{t+1} = \lambda_2^t$.

Suppose agent 1 chooses high effort, so $\sigma_1(\lambda_1^t, \lambda_2^t) = 1$. It follows from equation (1) that for any output y , $\lambda_1^{t+1}(y) = \lambda_1^t$. Intuitively, agent 1 is mimicking the reliable type behavior, which renders output uninformative. Subsequently, neither agent's reputation changes, so the state on which agents condition their effort choices remains constant. Each agent's effort choice in period $t + 1$ must be the same as in period t . Inductively, if $\sigma_1(\lambda_1^t, \lambda_2^t) = 1$, then for each agent i , $\sigma_i(\lambda_1^{t+k}, \lambda_2^{t+k}) = \sigma_i(\lambda_1^t, \lambda_2^t)$ for all $k \geq 0$.

We next show that a state in which one strategic agent exerts high effort while the other shirks cannot occur in equilibrium. Suppose instead that at some time t , agent 1 exerts high effort while agent 2 shirks. By the previous argument, these actions persist forever. Since the consumer is myopic, she strictly prefers hiring agent 1 in every subsequent period. However, if agent 1 knows that he will be hired in each subsequent period regardless of the history, then he has a profitable deviation. By deviating to shirking, he saves the effort cost without affecting future payoffs.

We conclude that if a strategic agent exerts high effort in some period t , then both strategic agents must exert high effort in every period $t' \geq t$. Therefore, any pure strategy equilibrium follows a simple pattern: the game starts with each strategic-type agent shirking. Since the consumer is myopic, she hires the agent with the higher reputation. This happens until some time τ , after which both agents choose high effort indefinitely regardless of type. The following proposition summarizes the structure of all pure strategy equilibria:

Proposition 1. *Let σ be a pure strategy equilibrium. There exists a (possibly history-dependent) time $\tau \in \mathbb{N} \cup \{\infty\}$ such that*

1. *For all $t < \tau$, $\sigma_1(\lambda_1^t, \lambda_2^t) = \sigma_2(\lambda_1^t, \lambda_2^t) = 0$ and $x_t \in \arg \max_i \lambda_i^t$.*
2. *For all $t \geq \tau$, $\sigma_1(\lambda_1^t, \lambda_2^t) = \sigma_2(\lambda_1^t, \lambda_2^t) = 1$.*

In the remainder of the paper, for any equilibrium σ , we refer to the first period of high effort as τ_σ . Since monitoring is imperfect, if $\tau_\sigma < \infty$, then both reputations permanently remain fixed at strictly positive values. As a result, reputations do not vanish. We show in Appendix B that if $\tau_\sigma = \infty$, then both reputations vanish almost surely if both agents are strategic. This permits the following proposition:

Proposition 2. *In any equilibrium σ , the probability that both agents' reputations vanish is strictly*

positive if and only if τ_σ equals infinity with positive probability.

Proposition 2 shows that reputations vanish if and only if the game never enters the working phase. The next section first studies when it is possible for play in the working phase to be sustained as an equilibrium. The following section provides a sufficient condition for τ_σ to be almost surely finite for any equilibrium σ .

5 Sustainability of Effort

Suppose there exists an equilibrium σ such that $\tau_\sigma < \infty$ with positive probability. By Proposition 1, both strategic agents exert high effort in every period $t \geq \tau_\sigma$. Thus, play in the continuation game coincides with the consumer's first-best outcome. It follows that the continuation strategies following τ_σ must themselves constitute a first-best equilibrium. Consequently, if no first-best equilibrium exists, then there can be no equilibrium for which $\tau_\sigma < \infty$ with positive probability. This section characterizes when a first-best equilibrium exists. First, a definition:

Definition 1. *An equilibrium σ is first-best if $\sigma_i(\lambda_1^t, \lambda_2^t) = 1$ for each player i and time $t \in \mathbb{N}$.*

Intuitively, the consumer incentivizes high effort by threatening to fire an agent if output is too low. For a first-best equilibrium to exist, the expected loss in future payoffs must outweigh the cost savings from a one-shot deviation to low effort. The following theorem provides a necessary and sufficient condition for this to hold.

Theorem 1. *There exists a first-best equilibrium if and only if there exists $\tilde{y} \in Y$ such that*

$$\frac{(1-c)(1-\delta+\delta F_1(\tilde{y}))}{1-\delta+2\delta F_1(\tilde{y})} \geq \frac{1-\delta+\delta F_0(\tilde{y})(1-c)}{1-\delta+2\delta F_0(\tilde{y})}.$$

The proof comes from a straightforward application of Abreu, Pearce, and Stacchetti (1990). To understand the condition provided in Theorem 1, a key observation of the proof is that if there exists a first-best equilibrium, then there exists one in which the consumer's strategy takes the following form:

1. The consumer starts by hiring agent 1.
2. In each period t , if the consumer hires agent i , she will hire agent i in period $t+1$ if and only if output exceeds $\tilde{y}$.

The first-best condition provided in Theorem 1 simply states that neither agent has a profitable one-shot deviation given the above consumer strategy. The left side of the inequality is the agent's continuation payoff in any subgame that starts with him being hired. The right side is his payoff from a one-shot deviation to low effort.

Example: To understand what Theorem 1 requires of the model primitives, we consider a simple example. Suppose output is binary: $Y = \{y_L, y_H\}$ with $y_L < y_H$. Suppose that the probability of achieving y_H under effort e is p_e , where $p_1 = 1 - p_0 > 1/2$. Theorem 1 states that a first-best

equilibrium exists if and only if $\frac{(1-c)(1-\delta+\delta p_1)}{1-\delta+2\delta p_1} \geq \frac{1-\delta+\delta p_0(1-c)}{1-\delta+2\delta p_0}$. Rearranging this inequality yields the following:

There exists a first-best equilibrium if and only if $\delta(p_1 - p_0) \geq c$.

From this, it is clear that for there to exist a first-best equilibrium, the cost of effort must not be too high and agents must be sufficiently patient. Finally, output must be sufficiently informative about effort. If high effort does not sufficiently improve the chances of achieving high output, then it is not incentive compatible for either agent to work hard indefinitely. These requirements hold in the general case.

Before moving on to the next section, we note the following:

Corollary 1. *If there exists a first-best equilibrium, then there exists a pure strategy Markov equilibrium.*

Proof. Let σ be a first-best equilibrium. By definition, each strategic agent exerts high effort in every period independent of the history. Hence each agent's strategy can be represented as the Markov strategy $\sigma_i(\lambda_1, \lambda_2) = 1$ for all $(\lambda_1, \lambda_2) \in [0, 1]^2$. Therefore, σ is a pure strategy Markov equilibrium. $\square$

Recall that any equilibrium that reaches the working phase in finite time must sustain permanent high effort thereafter. Consequently, a first-best equilibrium must exist for such an outcome to be feasible. The remainder of this paper assumes that a first-best equilibrium exists.

6 Long-run Reputations

Section 5 established that if there exists a first-best equilibrium, then there exists an equilibrium in which neither agent's reputation vanishes. We now turn to the stronger question of when reputations do not vanish in any equilibrium. We begin by defining an important object. For any equilibrium σ , there is a corresponding probability distribution over outcomes, which we denote by P_σ .⁵ Mailath and Samuelson (2006) show that λ_1^t and λ_2^t are P_σ martingales bounded in L^1 , so both converge almost surely.

We first show that it is impossible for one agent's reputation to vanish while the other agent's reputation survives.

Lemma 1. *For any equilibrium σ and agent i , $P_\sigma(\lim_{t \rightarrow \infty} \lambda_i^t = 0 \text{ and } \lim_{t \rightarrow \infty} \lambda_{-i}^t > 0) = 0$.*

Proof. Suppose, for contradiction, that:

⁵We formally define P_σ and the outcome space in Appendix A.

$$P_\sigma(\lim_{t \rightarrow \infty} \lambda_1^t = 0 \text{ and } \lim_{t \rightarrow \infty} \lambda_2^t > 0) > 0.$$

Consider any path along which λ_1^t converges to 0. Such a path possesses three properties. First, agent 1 must be hired infinitely often. If there exists $\tilde{t}$ such that the consumer hires agent 2 in every period $t' > \tilde{t}$. Then $\lambda_1^{t'} = \lambda_1^{\tilde{t}} > 0$. This contradicts that λ_1^t converges to 0.

Second, along any path in which agent 1's reputation converges to 0, he must shirk in every period in which he is hired. If agent 1 chooses high effort at any time $t < \infty$, his reputation will permanently remain fixed at $\lambda_1^t > 0$ (see Section 4).

Finally, if only agent 1's reputation converges to 0, then there must be some time T for which $\lambda_1^t < \lambda_2^t$ for all $t \geq T$. However, if agent 1 is hired infinitely often and shirks when hired, there exists $t \geq T$ such that agent 1 is hired in period t , and his total probability of high effort is $\lambda_1^t < \lambda_2^t$. The consumer has a profitable deviation by hiring agent 2 in period t , contradicting that σ is an equilibrium. $\square$

Lemma 1 implies that either both reputations vanish or neither does. The remainder of this section discusses when the former does not occur. By Proposition 2, this is equivalent to showing $P_\sigma(\tau_\sigma < \infty) = 1$ for any equilibrium σ . Our main result provides a sufficient condition for this to hold. Intuitively, such a condition should make deviations to high effort sufficiently attractive. The condition we offer is as follows:

Condition 1. $F_0(y_*) - F_1(y_*) > c$.

For any y , $F_0(y) - F_1(y)$ measures how much high effort improves the probability of output exceeding y . This value is maximized at y_* . Condition 1 simply states that the maximal probability improvement exceeds the cost of effort. We prove that if Condition 1 holds, then as long as agents are sufficiently patient, all equilibria almost surely enter the working phase in finite time.

Theorem 2. *Under Condition 1, there exists $\underline{\delta} < 1$ such that if $\delta \geq \underline{\delta}$, then for any equilibrium σ , $P_\sigma(\tau_\sigma < \infty) = 1$.*

For intuition, suppose σ is an equilibrium in which both agents shirk after every history, so that $\sigma_i(\lambda_1, \lambda_2) = 0$ for each agent i and reputation pair (λ_1, λ_2) . Then $\tau_\sigma = \infty$ almost surely, implying that both reputations vanish almost surely. Since the consumer's best response is to always hire the agent with the higher reputation, the incumbent remains employed only until his reputation falls below his rival's. As both reputations converge to zero, they eventually become so close that a small decline in the incumbent's reputation is enough to overturn the ranking. Condition 1 implies that a sufficiently patient incumbent has a profitable deviation to high effort, since increasing the probability of maintaining his reputational advantage outweighs the effort cost.

To see this, suppose agent 1 loses his reputational advantage if and only if output is below y_* . Let x denote his continuation payoff beginning two periods from now conditional on today's output being below y_* . So if agent 1 loses his lead, he gets payoff 0 tomorrow plus δx . Let x' be his continuation payoff in two periods conditional on today's output exceeding y_* . If he exerts low effort today, his payoff is

$$V_0 = (1 - \delta) + \delta(F_0(y_*)(0 + \delta x) + (1 - F_0(y_*))(1 - \delta + \delta x')).$$

If he deviates to high effort, his payoff is

$$V_1 = (1 - \delta)(1 - c) + \delta(F_1(y_*)(0 + \delta x) + (1 - F_1(y_*))(1 - \delta + \delta x')).$$

Comparing these two expressions tells us that agent 1 has a profitable deviation if and only if

$$(F_0(y_*) - F_1(y_*))(1 - \delta + \delta(x' - x)) > \frac{(1 - \delta)c}{\delta}.$$

In the appendix, we prove $x' - x$ must be positive. Intuitively, retaining a higher reputation improves future hiring prospects, so the continuation value from preserving one's lead exceeds the continuation value from losing it. Hence, deviating to high effort is profitable if

$$(F_0(y_*) - F_1(y_*))(1 - \delta) > \frac{(1 - \delta)c}{\delta}.$$

Under Condition 1, this inequality holds for δ sufficiently close to 1.

The preceding argument shows that under Condition 1, when agents are sufficiently patient, there is no equilibrium σ in which $\tau_\sigma = \infty$ almost surely. Theorem 2 strengthens this conclusion by ruling out equilibria in which $\tau_\sigma = \infty$ with positive probability. The key step is to show that if there is an equilibrium σ for which $P_\sigma(\tau_\sigma = \infty) > 0$, then with positive probability, the path of play reaches a continuation game in which $\tau_\sigma = \infty$ with probability arbitrarily close to 1. At this continuation game, the same deviation argument applies.

Recall that Cripps, Mailath, and Samuelson (2004) show that in a game between one long-lived player and one short-lived player, the long-lived player's reputation will dissipate over time. The intuition behind this result is that the only way the long-lived player can maintain a reputation as being committed to a particular action is by repeatedly choosing that action over time. However, since individual signals become less informative over time, deviating from the commitment action has negligible benefits on future payoffs and strictly improves current payoffs. In our setting, beliefs still converge, and individual signals become less informative. However, if both reputations

disappear, then eventually the two reputations become arbitrarily close. One more bad signal causes the agent with the higher reputation to lose his advantage, leading to a loss in payoffs of at least $1 - \delta$.

7 Extensions

7.1 Comment on independence

So far, we have assumed that output generated by one agent provides no information about his opponent's type. An important consequence of this is that an agent who shirks indefinitely will almost surely lose his reputational advantage.⁶ The argument behind Theorem 2 is that at some point before this happens, the agent will choose to deviate to high effort. However, depending on the degree of correlation between types, the incumbent may never face the risk of losing his advantage.

To illustrate, suppose that rather than agent's types being independently distributed, it is commonly known that the two agents have the same type: either both are strategic, or both are reliable. It is then trivial to verify that the following is an equilibrium: in every period, independent of the history, both strategic agents choose to shirk and the consumer hires agent 1. If agent 1 deviates to high effort, he incurs a current period cost with no benefit to future payoffs. Thus choosing high effort in any period is not profitable. The consumer also never has a profitable deviation because the probability of high effort is always the same across the two agents. If agent 1 shirks in every period (conditional on being strategic) and types are perfectly positively correlated, then both reputations almost surely converge to zero. Theorem 2 subsequently fails.

In this extreme example, agent types were perfectly positively correlated. Intuitively, positive correlation implies that if agent 1's reputation decreases, so will his opponent's. Therefore, in comparison to when types are independent, low output does not impose as severe of a penalty on an agent's advantage. This weakens incentives to work hard. On the other hand, if the correlation between types is negative, Theorem 2 is still true:

Proposition 3. *Assume agent types are negatively correlated and Condition 1 holds. There exists $\underline{\delta} < 1$ such that if $\delta \geq \underline{\delta}$, then in any equilibrium σ , $P_\sigma \left(\lim_{t \rightarrow \infty} \min\{\lambda_1^t, \lambda_2^t\} > 0 \right) = 1$.*

We do not include a proof in the appendix. Negatively correlated types implies that if one agent's reputation rises, his opponent's reputation falls. From this, one may directly repeat the arguments in the proof of Theorem 2.

⁶If an agent shirks indefinitely conditional on being strategic, his reputation converges to zero. If his opponent's reputation remains static, then the ranking of reputations will eventually flip.

7.2 First-best versus asymptotic first-best

It is natural to ask how Condition 1 relates to the condition implying the existence of a first-best equilibrium. Proposition 4 shows that the conditions are closely related.

Proposition 4. *1. If $1 - F_1(y_*) \geq \frac{1+\delta c}{2\delta}$, then Condition 1 implies the existence of a first-best equilibrium.*

2. If there exists $\tilde{y}$ such that a first-best equilibrium exists, and $\frac{1+\delta c}{2\delta} \geq 1 - F_1(\tilde{y})$, then Condition 1 is satisfied.

Proposition 4 shows that the condition guaranteeing non-vanishing reputations is closely related to the condition guaranteeing the existence of a first-best equilibrium. Importantly, our finding that all equilibria attain eventual first-best does not preclude the existence of a first-best equilibrium. This differs from Kartik, Lipnowski, and Pei (2025), who show that the existence of a first-best equilibrium is incompatible with all equilibria achieving eventual first-best. The difference stems from the equilibrium restrictions. Their analysis allows for mixed strategies, whereas our results are derived within the class of pure Markov equilibria. It is possible that equilibria outside this class fail to attain eventual first-best, but studying such equilibria is beyond the scope of the paper.

8 Conclusion

This paper studies the long-run consequences of competition in reputation building. We show that competition can prevent reputations from vanishing, even when reputations would disappear in a comparable environment with a single long-lived agent. In particular, we characterize conditions under which a first-best equilibrium exists and show that, under a related condition, reputations never vanish in any pure Markov equilibrium.

We contrast this result with the seminal result of Cripps, Mailath, and Samuelson (2004). In our game, high effort is not possible in any Nash equilibrium of the stage game with complete information. Therefore, if there were only one agent, then in any Nash equilibrium of the repeated game with incomplete information, his reputation would vanish. Competition fundamentally changes the agents' incentives. Even if absolute reputations fall, the benefit of having a higher reputation than one's opponent is enough for an agent to permanently invest in his reputation.

We conclude with two interesting future directions.

First, we assume that the consumer may freely switch back and forth between agents. This is intuitive but not universally plausible. For example, Kartik, Lipnowski, and Pei (2025) assume that agents who are fired may never be rehired. One may alternatively imagine a scenario where the consumer faces switching costs. Such modifications may make it more difficult for the consumer to incentivize high effort. In the latter case, an agent may feel stronger incentives to shirk if he knows the consumer is less willing to fire him.

Finally, we assume that neither agent knows his opponent's type. However, it is likely that an agent who is competing against a strategic opponent will behave differently than if he knows that his opponent is reliable. An interesting question is how this influences the agent's behavior and to what extent the consumer can learn about one agent's type by hiring his opponent.

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A Preliminaries

Here we formally define strategies and provide useful notations that will be used in Appendix B.

A.1 Formal Definition of strategies and spaces

We start by defining the histories and strategies of each of the players.

Definition 2. *The set of period t histories is given by the set $H_t := (\{1, 2\} \times Y)^t$. The consumer's period t strategy is a map $\sigma_c : \cup_{t \geq 0} H_t \rightarrow \{1, 2\}$.*

Histories are observable to all players. We now define the agent strategies. Since the agent's strategy is only relevant when he is strategic, we denote by σ_i the agent's strategy conditional on being strategic. Agent i 's strategy is a map $\sigma_i : \cup_t H_t \rightarrow \{0, 1\}$ specifying an effort choice based on the agent's type as well as the period t history. We impose the following restriction on agent strategies:

Definition 3. *Agent i 's strategy is **Markov** if $\forall h_t \in H_t, h_{t'} \in H_{t'}$, if $(\lambda_1^t(h_t), \lambda_2^t(h_t)) = (\lambda_1^t(h_{t'}), \lambda_2^t(h_{t'}))$, then $\sigma_i(h_t) = \sigma_i(h_{t'})$.*

For any period t history, let $v_i(h_t)$ denote player i 's payoff in the continuation game starting at h_t . We now define an equilibrium as follows:

Definition 4. *A perfect Bayesian equilibrium is a triple of strategies $(\sigma_c, \sigma_1, \sigma_2)$ as well as a conditional belief system $\lambda : \cup_t H_t \rightarrow [0, 1]^2$ such that:*

1. $\forall h_t \in H_t, t \geq 0, \sigma_c(h_t) \in \arg \max_{i \in \{1, 2\}} \lambda_i(h_t) + (1 - \lambda_i(h_t))\sigma_i(h_t).$
2. $\forall h_t \in H_t, i = 1, 2, \sigma_i(h_t) \in \arg \max_e (1 - \delta)(1 - c_e) + \mathbb{E}_e[v_i(h_t, (i, y_t))].$
3. $\forall h_t \in H_t, i = 1, 2, \lambda_i(h_t) = \frac{\lambda_i^0 \Pi_{t' < t: \sigma_c(h_{t'})=i} f_1(y_{t'})}{\lambda_i^0 \Pi_{t' < t: \sigma_c(h_{t'})=i} f_1(y_{t'}) + (1 - \lambda_i^0) \Pi_{t' < t: \sigma_c(h_{t'})=i} f_{\sigma_i(h_{t'})}(y_{t'})}.$
4. *Agent strategies are Markov.*

Any triple of strategies, σ will induce a probability distribution over outcomes. To define this formally, let $T = \{rr, rs, sr, ss\}$ be the set of type-pairs of the two agents, where element xy denotes that agent 1 is type x and agent 2 is type y . An agent's type being r implies he is reliable whereas s denotes the agent is strategic. Now we are ready to define an outcome:

Definition 5. *An outcome is an element of $\Omega := T \times (\{1, 2\} \times Y)^\mathbb{N}$ specifying agent types as well as an infinite sequence of agent selections and output levels.*

Assume Ω is measurable with some sigma algebra $\mathcal{A}$. Any triple of strategies $\sigma = (\sigma_c, \sigma_1, \sigma_2)$ induces a probability measure P_σ over Ω . We now define agent reputations. For any event $F \in \mathcal{A}$, let $\mathbb{1}\{F\} = 1$ if $\omega \in F$ and 0 otherwise.

Let $\mathcal{H}_t$ be the sigma algebra induced by H_t . $(\mathcal{H}_t)_{t \in \mathbb{N}}$ is a filtration on Ω . For $i = 1, 2$, define agent i 's **period t reputation** to be given by the $\mathcal{H}_t$ -measurable random variable $\lambda_i^t := E_\sigma[\mathbb{1}\{i \text{ is reliable}\} | \mathcal{H}_t]$. In an abuse of notation, for any $h_t \in H_t$, we let $\lambda_i^t(h_t) = E_\sigma[\mathbb{1}\{i \text{ is reliable}\} | h_t]$.

If both strategies are Markov, it is without loss to express agent strategies as a map $\sigma_i : [0, 1]^2 \rightarrow \{0, 1\}$ specifying the effort choice of agent i given the pair of agent reputations. If agent i 's type is r , he plays the strategy: $\tilde{\sigma}$, where for any reputation pair (λ_1, λ_2) , $\sigma(\lambda_1, \tilde{\lambda}_2) = 1$.

Then given initial reputations λ_1, λ_2 and strategy profile σ , we can express the probability distribution P_σ as follows:

$$P_\sigma = \lambda_1 \lambda_2 P_{(\sigma_c, \tilde{\sigma}, \tilde{\sigma})} + \lambda_1 (1 - \lambda_2) P_{(\sigma_c, \tilde{\sigma}, \sigma_2)} + (1 - \lambda_1) \lambda_2 P_{(\sigma_c, \sigma_1, \tilde{\sigma})} + (1 - \lambda_1) (1 - \lambda_2) P_{(\sigma_c, \sigma_1, \sigma_2)}.$$

Lemma 2. λ_i^t is a martingale bounded in L_1 and therefore converges P_σ almost surely.

See Mailath and Samuelson (2006) for a proof.

A.2 Notation

We introduce some basic notation that will be useful in the remainder of the appendix.

- For any vector $v \in \mathbb{R}^3$, we specify the vector as (v_c, v_1, v_2) where v_c specifies the consumer's component, and v_i is agent i 's component.
- For any random variable g whose value is a function of the output y , we denote $\mathbb{E}_1[g(y)]$ as the expected value of g when the agent's effort is equal to 1, and $\mathbb{E}_0[g(y)]$ if the agent's effort is 0.
- For any set $C \subseteq \mathbb{R}^n$, let $co(C)$ be the convex hull of C
- For any set $A \subseteq \mathbb{R}$, the set $1 - A = \{1 - a : a \in A\}$

B Proofs of Theorem 1 and Theorem 2

B.1 Proof of Theorem 1

First, let Γ_{inc} be the game of incomplete information as defined in the model. We prove Theorem 1 by showing existence in an auxiliary game. Let Γ_{com} denote the game of complete information in which both agents are commonly known to be strategic. A first-best equilibrium in Γ_{com} is a subgame perfect equilibrium σ such that $\sigma_1(h_t) = \sigma_2(h_t) = 1 \forall h_t \in H_t, t \geq 0$.

B.1.1 Step 1: Equivalence to Complete Information Game

Claim 1. *There exists a first-best equilibrium iff there exists a first-best equilibrium in Γ_{com}*

Proof. Suppose there exists a first-best equilibrium σ^* in Γ_{inc} . We show that σ^* is also an equilibrium in Γ_{com} . Since $\sigma_i^*(h_t) = 1$ for all h_t , the consumer is indifferent between choosing agent 1 and agent 2 at each history. Therefore, σ_c^* is a best response to (σ_1^*, σ_2^*) in Γ_{com} .

Now we partition the set of histories into two sets: $H^1 := \{h_t \in H_t : \sigma_c^*(h_t) = 1\}$ and $H^2 := \{h_t : \sigma_c^*(h_t) = 2\}$. In either game, the agent's flow payoff under strategy profile σ^* for any history in H^1 equals $1 - c$ and for any history in H^2 equals 0. We will show that for any history $h_t \in H^1$, agent 1 has no profitable one shot-deviation in Γ_{com} . Let $(h_t, (1, y)) \in H_{t+1}$ be a succeeding history. The probability of reaching this history starting from history h_t equals $f_1(y)$. This is true in both Γ_{com} and Γ_{inc} . Furthermore, for any history $h_t \in H^2$, the probability of reaching history $(h_t, (2, y))$ equals $f_1(y)$ in Γ_{com} and $\lambda_2(h_t)f_1(y) + (1 - \lambda_2(h_t))f_{\sigma_2^*(h_t)}(y) = f_1(y)$ in Γ_{inc} . These two probabilities are the same.

Now suppose that for $k \geq 1$, the probability of reaching history $h_{t+k} \in H_{t+k}$ starting from h_t is the same in Γ_{com} as it is in Γ_{inc} . We show that for any history $h_{t+k+1} \in H_{t+k+1}$, the probability of reaching h_{t+k+1} starting from h_t is the same in Γ_{com} as it is in Γ_{inc} . Let $h_{t+k+1} = (h_{t+k}, (1, y))$. Then the probability of reaching h_{t+k+1} starting from h_t in Γ_{com} equals $P(h_{t+k}|h_t)f_1(y)$. From the inductive step, the probability of reaching history h_{t+k} is the same as in Γ_{inc} . Since the probability of reaching history h_{t+k+1} from history h_{t+k} is the same in both games, we conclude that the probability of reaching history h_{t+k+1} starting from history h_t is the same in both games. The same argument applies if $h_{t+k+1} = (h_{t+k}, (2, y))$.

We conclude that if the flow payoffs as well as the transition probabilities between histories are the same in Γ_{com} and Γ_{inc} , then for any $h_t \in H^1$, agent 1 cannot have a profitable one-shot deviation in Γ_{com} if σ^* is an equilibrium of Γ_{inc} . Otherwise, he would have a profitable one-shot deviation in Γ_{inc} , which contradicts σ^* being an equilibrium. The same logic applies to why agent 2 has no profitable one-shot deviation.

The argument for the opposite direction is exactly the same and the proof is complete. $\square$

By Claim 1, proving Theorem 1 is equivalent to proving existence of a first-best equilibrium in Γ_{com} . In any period t , assuming pure strategies, the vector of stage game payoffs will be equal to one of $(1, 1 - c, 0)$, $(1, 0, 1 - c)$, $(\underline{y}, 1, 0)$, $(\underline{y}, 0, 1)$. For a strategy profile σ in Γ_{inc} , let $v(\sigma)$ be the ex-ante payoff vector to the three players. Then let $V = \{v(\sigma) : \sigma \text{ is an SPE}\}$. It follows that:

$$V \subset \text{co}\{(1, 1 - c, 0), (1, 0, 1 - c), (\underline{y}, 1, 0), (\underline{y}, 0, 1)\}$$

Now let $\bar{V} = \{v(\sigma) : \sigma \text{ is a first-best equilibrium}\}$. Define the set C to be $\text{co}\{(1, 1 - c, 0), (1, 0, 1 - c)\}$.

Claim 2. $\bar{V} = V \cap C$

Proof. ($\subseteq$) Let $v \in \bar{V}$. Since, $\bar{V} \subset V$, it is trivial that $v \in V$. It remains to show that $v \in C$. That

$v \in \bar{V}$ implies $v_c = 1$. Therefore, since $V \subset \text{co}\{(1, 1 - c, 0), (1, 0, 1 - c), (\underline{y}, 1, 0), (\underline{y}, 0, 1)\}$, it must be that $v \in \text{co}\{(1, 1 - c, 0), (1, 0, 1 - c)\} = C$.

($\supseteq$) Let $v \in V \cap C$. Then there exists a subgame perfect equilibrium σ such that for each period t , $\sigma_{i_t}(h_t) = 1$ almost surely, where i_t is the agent hired in period t . For any h_t such that $\sigma_{i_t}(h_t) < 1$, we may rewrite the agent's strategy profile in the continuation game to equal the original game's strategy profile. Doing this infinitely often gives a subgame perfect equilibrium with high effort at every history. That it is subgame perfect follows because the strategy adjustment occurs for a measure zero set of histories and therefore does not alter incentives.

□

Claim 2 simply states that the set of possible first-best equilibrium payoff vectors is equal to the total set of subgame perfect equilibrium payoff vectors intersected with the set C . The next step is to show that proving existence of a first-best equilibrium in Γ_{com} is equivalent to showing the existence of a non-empty self-generating subset of C . Before proceeding, we define the APS operator as well as self-generation:

- Let (q, u) be a pair, where $q \in \{1, 2\} \times \{0, 1\}^2$ is an action profile in the stage game, and u is a function from $\{1, 2\} \times Y$ to $\mathbb{R}^3$ that specifies the continuation payoffs to each player after the public realization of the agent selected and output.
- We define $E(q, u) := (1 - \delta)(E_{q_c}[y], \mathbb{1}\{q_c = 1\}(1 - cq_1), (\mathbb{1}\{q_c = 2\})(1 - cq_2)) + \delta E_{q_c}[u(q_c, y)]$. This is simply the weighted average between today's stage game payoff vector and expected future payoffs.

We now define admissibility according to the standard definition in APS.

Definition 6. A pair (q, u) is admissible with respect to a set $W \subseteq \mathbb{R}^3$ if $u(i, y) \in W \forall i = 1, 2, y \in Y$ and the following incentive conditions hold:

$$\bullet q_c \in \arg \max_{i \in \{1, 2\}} E_{q_i}[y] \tag{IC1}$$

$$\bullet \text{ If } q_c = i, \text{ then } q_i = 1 \text{ iff } (1 - \delta)(1 - c) + \delta E_1[u_i(i, y)] \geq (1 - \delta) + \delta E_0[u_i(i, y)] \tag{IC2}$$

The APS operator is given by $B(W) = \{w : w = E(q, u) \text{ for } (q, u) \text{ admissible wrt } W\}$. I provide the standard definition of a self-generating set:

Definition 7. A bounded set W is self-generating if $W \subseteq B(W)$.

If a bounded set W is self-generating, then $B(W) \subseteq V$. This is a standard result from APS.

B.1.2 Step 2: Proof via Self-Generation

By APS Theorem 4, V is compact. Since $\bar{V}$ is the intersection of two compact sets in $\mathbb{R}^3$, $\bar{V}$ is also compact. The next lemma states that there exists a first-best equilibrium in Γ_{com} only if there exists a non-empty self-generating subset of C .

Lemma 3. *If there exists a non-empty, bounded self-generating set $W \subset C$, then $\bar{V}$ is non-empty and $W \subseteq \bar{V}$. Furthermore, if there exists a non-empty, bounded self-generating set W such that $B(W) \cap C \neq \emptyset$, then $W \cap C \neq \emptyset$ and $W \cap C \subset B(W) \cap C \subset B(W \cap C)$.*

Proof. Let $W \subseteq C$ be self-generating. Then $W \subseteq B(W) \subseteq V$, which implies $W \subseteq V \cap C = \bar{V}$.

Suppose that for some self-generating set W , that $B(W) \cap C \neq \emptyset$ but $W \cap C = \emptyset$. Let $v \in B(W) \cap C$. It is clear that $v_c = 1$. Note, though, that if $W \subseteq V$, then $W \subseteq \text{co}\{(1, 1-c, 0), (1, 0, 1-c), (y, 1, 0), (y, 0, 1)\}$. So then if q, u is admissible with respect to W , then $u_c(i, y) = 1$ for each agent i and output level y . This contradicts that $W \cap C = \emptyset$. Furthermore, (q, u) must be admissible with respect to $W \cap C$. This is simply because for all $w \in W \setminus C$, $w_c < 1$. Thus, $B(W) \cap C \subseteq B(W \cap C)$. $\square$

As a direct corollary of Lemma 3, we get that:

Lemma 4. $\bar{V} = B(\bar{V})$

This result tells us that there exists a first-best equilibrium iff there exists a nonempty self-generating subset of C . we now show an important symmetry property of any self-generating subset of C .

B.1.3 Step 3: Important properties of any self-generating subset of C

The next claim justifies that it is without loss to restrict attention to symmetric self-generating subsets of C .

Claim 3. *Suppose $W \subset C$ is self-generating. Then the set $W' = \{(w_c, w_2, w_1) : (w_c, w_1, w_2) \in W\}$ is also self-generating.*

Proof. Suppose $W \subset C$ is self-generating. Then for any $w \in W$, there exists $q \in \{1, 2\} \times \{0, 1\}^2$ and $u : \{1, 2\} \times Y \rightarrow W$ satisfying (IC1) and (IC2). Suppose without loss of generality that $q_c = 1$. Then

1. $w_1 = (1 - \delta)(1 - c) + \delta E_1[u_1(1, y)]$
2. $w_2 = \delta E_1[u_2(1, y)]$

Now define w' as the vector obtained by switching the payoffs of agents 1 and agent 2: $(w'_c, w'_1, w'_2) = (w_c, w_2, w_1)$. We wish to find a pair q', u' admissible with respect to W' such that $E(q', u') = w'$. Define $q' = (2, 1, 1) \in \{1, 2\} \times \{0, 1\}^2$, and the function $u' : \{1, 2\} \times Y \rightarrow \mathbb{R}^3$ as follows:

- $u_c(i, y) = 1$
- $u'_1(2, y) = u_2(1, y) \forall y$
- $u'_2(2, y) = u_1(1, y) \forall y$

- $u'_1(1, y) = u_2(2, y)$ and $u'_2(1, y) = u_1(2, y)$

It is clear that $u'(i, y) \in W'$ for all values $i, y \in \{1, 2\} \times Y$. Given the construction, (q', u') is admissible with respect to W' if and only if (q, u) is admissible with respect to W , which is true by assumption. This completes the proof. $\square$

Claim 3 essentially tells us that we can without loss think about symmetric self-generating subsets of C . Therefore, going forward, whenever we discuss a set $W \subset C$, we assume that if $w = (w_c, w_1, w_2) \in W$, then $(w_c, w_2, w_1) \in W$ as well.

B.1.4 Step 4: One Dimensional reduced problem

Here we show that we may reduce the problem of finding a self-generating subset of C to a simpler one-dimensional problem. Consider a compact set $W \subset C$. For any $w \in W$, $w = (1, \alpha(1 - c), (1 - \alpha)(1 - c))$ for some $\alpha \in [0, 1]$. Let $\alpha(W) = \{\alpha \in [0, 1] : (1, \alpha(1 - c), (1 - \alpha)(1 - c)) \in W\}$. By claim 3, if $a \in \alpha(W)$ then $1 - a \in \alpha(W)$. The next step of the proof is to essentially redefine the set of equilibrium payoffs into the set of weights compatible with equilibrium payoffs for the two players.

Claim 4. *$W \subset C$ is self-generating if and only if $\forall a \in \alpha(W)$, there exists a function $u : Y \rightarrow \alpha(W)$ such that exactly one of the following two conditions hold:*

- $a = (1 - \delta) + \delta E_1[u(y)]$ and $a(1 - c) \geq (1 - \delta) + \delta(1 - c)E_0[u(y)]$ OR
- $1 - a = (1 - \delta) + \delta E_1[u(y)]$ and $(1 - a)(1 - c) \geq (1 - \delta) + \delta(1 - c)E_0[u(y)]$

It is at this point of the proof that we appeal to compactness of $\bar{V}$.

Claim 5. *Suppose $W \subset C$ is compact and self-generating. Then there exists $\bar{\alpha} \in \alpha(W)$ such that $\bar{\alpha} \geq \alpha \geq 1 - \bar{\alpha}$ for all $\alpha \in \alpha(W)$, and $1 - \bar{\alpha} \in \alpha(W)$.*

Proof. If W is compact, then $\alpha(W)$ is also compact. That $\alpha(W)$ is bounded is obvious since $\alpha(W) \subset [0, 1]$. That it is closed is also clear: suppose $(\alpha_n)_n$ is a sequence of values in $\alpha(W)$ such that $\alpha_n \rightarrow \alpha$. For each n , $w_n := (1, \alpha_n(1 - c), (1 - \alpha_n)(1 - c)) \in W$. It is clear that w_n converges to $(1, \alpha(1 - c), (1 - \alpha)(1 - c)) \in W$ since W is closed. Thus, $\alpha \in \alpha(W)$. Since $\alpha(W)$ is compact, there exists a maximal and minimal element. That they sum to 1 is a consequence of symmetry of W . $\square$

Now we use the bang-bang result from Abreu, Pearce, and Stacchetti (1990). If W is compact and self-generating, then for all $w \in W$, there exists an admissible pair (q, u) where $q \in \{1, 2\} \times \{0, 1\}^2$ and $u : \{1, 2\} \times Y \rightarrow \text{ext}(W)$ ⁷ such that $E(q, u) = w$.

For a compact self-generating set $W \subset C$, we can explicitly write $\text{ext}(W) = \{(1, \bar{\alpha}(1 - c), (1 - \bar{\alpha})(1 - c)), (1, (1 - \bar{\alpha})(1 - c), \bar{\alpha}(1 - c)), \bar{\alpha} = \max \alpha(W)\}$. Since $\text{ext}(W) \subset W$ for any self-generating compact subset of W , we can basically restrict attention to a two-point set.

⁷Here, as in APS, $\text{ext}(W)$ gives the set of extreme points of the convex hull of W .

Claim 6. *There exists a nonempty, compact, self-generating set $W \subset C$ if and only if there exists a function $u : Y \rightarrow \{\bar{\alpha}, 1 - \bar{\alpha}\}$ such that $\bar{\alpha} = 1 - \delta + \delta E_1[u(y)]$ and $\bar{\alpha}(1 - c) \geq (1 - \delta) + (1 - c)\delta E_0[u(y)]$ for some $\bar{\alpha} \in [0, 1]$*

Proof. ($\Rightarrow$) Follows from Theorem 3 of Abreu, Pearce, and Stacchetti (1990)

($\Leftarrow$) Let $W = \{(1, \alpha(1 - c), (1 - \alpha)(1 - c)), (1, (1 - \alpha)(1 - c), \alpha(1 - c))\}$, and define $q = (1, 1, 1)$. Clear that (q, u) is admissible with respect to W , which proves that W is self-generating. $\square$

To complete the proof, we simply must show that there exists some value $\alpha \in [0, 1]$ for which there exists a function $u : Y \rightarrow \{\alpha, 1 - \alpha\}$ such that $\alpha = (1 - \delta) + \delta E_1[u(y)]$ and $(1 - c)\alpha \geq (1 - \delta) + \delta(1 - c)E_0[u(y)]$.

For such a hypothetical α , let $A_+ = \{y : u(y) = \alpha\}$, and $A_- = Y/A_+$. We then have

$$\alpha = (1 - \delta) + \delta[F_1(A_+)\alpha + (1 - F_1(A_+))(1 - \alpha)]$$

Therefore,

$$\alpha = \frac{1 - \delta F_1(A_+)}{1 + \delta - 2\delta F_1(A_+)}$$

The IC constraint tells us that

$$\alpha(1 - c) \geq 1 - \delta + \delta(F_0(A_+)\alpha + (1 - F_0(A_+))(1 - \alpha))(1 - c) \Leftrightarrow (F_1(A_+) - F_0(A_+))(2\alpha - 1) \geq \frac{c(1 - \delta)}{\delta(1 - c)}$$

Plugging our expression for α into the last inequality gives us the following expression:

$$(F_1(A_+) - F_0(A_+)) \frac{1 - \delta}{1 + \delta - 2\delta F_1(A_+)} \geq \frac{c(1 - \delta)}{\delta(1 - c)}$$

Simplifying this expression, we get that there exists a first-best equilibrium iff there exists a subset $A \subset Y$ such that

$$\frac{1 + c}{1 - c} F_1(A) - F_0(A) \geq \frac{(1 + \delta)c}{(1 - c)\delta}$$

By Assumption 1, this is satisfied if and only if there exists a set A of the form $[y, \bar{y}]$ satisfying the above inequality. This completes the proof.

B.2 Proof of Theorem 2

Let σ be an equilibrium strategy profile. σ defines a Markov chain over the pairs of agent reputations. To elaborate, let $[0, 1]^2$ be the set of all possible reputation pairs. For any pair (λ_1, λ_2) , let $p(\cdot|\lambda_1, \lambda_2)$ be the unconditional transition density function between states. Formally, the initial state of the chain is $(\lambda_1^0, \lambda_2^0)$, and the transition density function is given by:

$$p(\lambda'_1, \lambda'_2|\lambda_1, \lambda_2) = \begin{cases} 1 & \lambda'_i = \lambda_i \text{ and } \sigma_i(\lambda_1, \lambda_2) = 1 \forall i \\ \lambda_i f_1(y) + (1 - \lambda_i) f_0(y) & \lambda'_i = \frac{\lambda_i f_1(y)}{\lambda_i f_1(y) + (1 - \lambda_i) f_0(y)}, \lambda'_{-i} = \lambda_{-i}, \lambda_i \geq \lambda_{-i}, \text{ and} \\ & \sigma_i(\lambda_1, \lambda_2) = \sigma_{-i}(\lambda_1, \lambda_2) = 0 \end{cases}$$

From the transition density function, we may define a probability measure $P_\sigma(\cdot|\lambda_1, \lambda_2)$ over outcomes that depends on the current reputations. For notational simplicity, we denote the probability measure given the initial state by $P_\sigma(\cdot)$.

We can analogously define the state-contingent probability measure over outcomes conditional on both agents being strategic. Define this measure by $\tilde{P}_\sigma(\cdot|\lambda_1, \lambda_2)$. Similar to the unconditional case, we denote the probability conditional on the initial state by $\tilde{P}_\sigma$.

One final remark is that $\tilde{P}_\sigma(\cdot|\lambda_1, \lambda_2)$ is absolutely continuous with respect to $P_\sigma(\cdot|\lambda_1, \lambda_2)$ for all reputation pairs (λ_1, λ_2) .

The transition densities allow for two possibilities. One possibility is that both agents shirk conditional on being strategic, and the agent hired has his reputation updated via Bayes' rule, whereas the agent not hired sees no change to his reputation. The other possibility is that the agent hired is exerting effort, in which case output is uninformative and reputations do not change. This corresponds to the set of absorbing states of the Markov chain. Let $\tau_\sigma = \inf\{t : \sigma_i(\lambda_1^t, \lambda_2^t) = 1 \text{ for some agent } i\}$. This is the first time at which an agent works hard (i.e, the time at which the Markov chain reaches an absorbing state).

We begin by showing that if $\tau_\sigma = \infty$, then λ_1^t and λ_2^t converge to 0 $\tilde{P}$ -almost surely and vice versa.

Lemma 5. *The events $\{\tau_\sigma = \infty\}$ and $\{\lambda_1^t, \lambda_2^t \rightarrow 0\}$ differ on a set N such that $\tilde{P}_\sigma(N) = 0$.⁸*

Proof. ($\subseteq$) Suppose $\tau_\sigma = \infty$. There exists i such that $\sigma_i(\lambda_1^t, \lambda_2^t) = i$ infinitely often. Suppose $i = 1$. By Bayes' rule,

$$\lambda_1^{t+1}(y_t) = \frac{\lambda_1^t f_1(y_t)}{\lambda_1^t f_1(y_t) + (1 - \lambda_1^t) f_0(y_t)}$$

Therefore

⁸The proof is also a proof of proposition 2 in section 4.

$$\frac{\lambda_1^{t+1}}{\lambda_1^t}(\lambda_1^t f_1(y_t) + (1 - \lambda_1^t) f_0(y_t)) = f_1(y_t). \quad (2)$$

Repeating the same computation with $1 - \lambda_i^{t+1}$ yields the following:

$$\frac{1 - \lambda_1^{t+1}}{1 - \lambda_1^t}((\lambda_1^t f_1(y_t) + (1 - \lambda_1^t) f_0(y_t))) = f_0(y_t). \quad (3)$$

By continuity of f_0, f_1 , we know that $\lambda_1^t f_1(y) + (1 - \lambda_1^t) f_0(y) < \max_{a,y} f_a(y) =: A < \infty$. Taking the absolute difference of (2) and (3) and using A yields the following inequality:

$$A \left| \frac{\lambda_1^{t+1}}{\lambda_1^t} - \frac{1 - \lambda_1^{t+1}}{1 - \lambda_1^t} \right| \geq |f_1(y_t) - f_0(y_t)|.$$

This inequality can be rearranged to obtain

$$A |\lambda_1^{t+1} - \lambda_1^t| \geq \lambda_1^t (1 - \lambda_1^t) |f_1(y_t) - f_0(y_t)|.$$

That λ_1^t is a P_σ -martingale implies it is almost surely convergent. Since $\tilde{P}_\sigma$ is absolutely continuous with respect to P_σ , λ_1^t converges $\tilde{P}_\sigma$ almost surely and thus is $\tilde{P}_\sigma$ almost surely Cauchy. The right hand side of the previous inequality therefore converges to 0 $\tilde{P}_\sigma$ almost surely. Since $f_1(y) = f_0(y)$ on a measure zero set, it must be that $\lambda_1^t (1 - \lambda_1^t)$ converges to 0 $\tilde{P}_\sigma$ almost surely. So λ_1^t converges either to 0 or 1. By Mailath and Samuelson (2006), λ_1^t converges to 1 with probability 0. Thus, λ_1^t must converge to 0 almost surely.

So if $\tau_\sigma = \infty$ and an agent is hired infinitely often, his reputation converges to 0. The last step is to show that both agents will be hired infinitely often. Suppose agent 2 is hired finitely often. Then there exists T s.t. $\forall t \geq T$, $\sigma_c(\lambda_1^t, \lambda_2^t) = 1$ and $\lambda_2^t = \lambda_2^T > 0$. This follows from independence of types. If λ_1^t converges to 0, then there exists $t' > T$ such that $\lambda_1^{t'} < \lambda_2^T$. If $\tau_\sigma = \infty$, $\sigma_c(\lambda_1^{t'}, \lambda_2^{t'}) = 1$ and $\lambda_1^{t'} < \lambda_2^{t'}$. The consumer then has a profitable deviation to hire agent 2, which contradicts that σ is an equilibrium.

($\supseteq$) Suppose $\tau_\sigma = T < \infty$. Then for all $t \geq T$, $\lambda_i^{t+1} = \lambda_i^t > 0$ for each $i = 1, 2$. Then $\lim_{t \rightarrow \infty} \lambda_i^t > 0$ for each i . $\square$

Mailath and Samuelson (2006) prove that if either agent is strategic, then his reputation almost surely does not vanish. Therefore, we deduce that $P_\sigma(\lambda_1^t, \lambda_2^t \rightarrow 0) > 0$ if and only if $\tilde{P}_\sigma(\lambda_1^t, \lambda_2^t \rightarrow 0) > 0$. Lemma 5 then tells us that $\tilde{P}_\sigma(\lambda_1^t, \lambda_2^t \rightarrow 0) > 0$ if and only if $\tilde{P}_\sigma(\tau_\sigma = \infty) > 0$.

We prove Theorem 2 by contradiction. Suppose that $P_\sigma(\tau_\sigma = \infty) > 0$. The next lemma argues that if the agents are strategic, then there must be a pair of reputations close to 0 at which the

consumer assigns a high probability to agents shirking forever.

Lemma 6. *For all $\alpha, \epsilon > 0$, there exists $(\lambda_1, \lambda_2) \in [0, \alpha]^2$ such that $\tilde{P}_\sigma(\tau_\sigma = \infty | \lambda_1, \lambda_2) > (1 - \epsilon)$.*

Proof. We prove this via contradiction. Suppose there exist $\alpha, \epsilon > 0$ such that $\forall (\lambda_1, \lambda_2) \in [0, \alpha]^2$, $\tilde{P}_\sigma(\tau_\sigma < \infty | \lambda_1, \lambda_2) \geq \epsilon$. Define the following stopping time:

$$T(\alpha) := \inf\{t : \max\{\lambda_1^t, \lambda_2^t\} < \alpha\}$$

This is the first time that both agent reputations fall below α . If $T(\alpha) = \infty$, then at least one of the agents' reputations never falls below α , and hence does not converge to 0.

Consider arbitrary reputations $\tilde{\lambda}_1, \tilde{\lambda}_2 \in (0, 1)^2$. It is without loss to consider interior reputation pairs since monitoring is imperfect. By the law of total probability, we decompose the probability that τ_σ is finite as follows:

$$\tilde{P}_\sigma(\tau_\sigma < \infty | \tilde{\lambda}_1, \tilde{\lambda}_2) = \tilde{P}_\sigma(\tau_\sigma < \infty, T(\alpha) < \infty | \tilde{\lambda}_1, \tilde{\lambda}_2) + \tilde{P}_\sigma(\tau_\sigma < \infty, T(\alpha) = \infty | \tilde{\lambda}_1, \tilde{\lambda}_2). \quad (4)$$

Consider the first term:

$$\tilde{P}_\sigma(\tau_\sigma < \infty, T(\alpha) < \infty | \tilde{\lambda}_1, \tilde{\lambda}_2) = \tilde{P}_\sigma(\tau_\sigma < \infty | T(\alpha) < \infty; \tilde{\lambda}_1, \tilde{\lambda}_2) \tilde{P}_\sigma(T(\alpha) < \infty | \tilde{\lambda}_1, \tilde{\lambda}_2)$$

By assumption and the Markov property, if we know $T(\alpha) < \infty$, then the probability that $\tau_\sigma < \infty$ is at least ϵ . Therefore

$$\tilde{P}_\sigma(\tau_\sigma < \infty, T(\alpha) < \infty | \tilde{\lambda}_1, \tilde{\lambda}_2) \geq \epsilon \tilde{P}_\sigma(T(\alpha) < \infty | \tilde{\lambda}_1, \tilde{\lambda}_2).$$

Now consider the second term:

$$\tilde{P}_\sigma(\tau_\sigma < \infty, T(\alpha) = \infty | \tilde{\lambda}_1, \tilde{\lambda}_2) = \tilde{P}_\sigma(\tau_\sigma < \infty | T(\alpha) = \infty; \tilde{\lambda}_1, \tilde{\lambda}_2) \tilde{P}_\sigma(T(\alpha) = \infty | \tilde{\lambda}_1, \tilde{\lambda}_2)$$

By Lemma 5, conditional on agents being strategic, $\tau_\sigma < \infty$ and reputations not vanishing differ on a null set. If $T(\alpha) = \infty$, then at least one agent's reputation does not converge to 0. As a result, $\tilde{P}_\sigma(\tau_\sigma < \infty | T(\alpha) = \infty; \tilde{\lambda}_1, \tilde{\lambda}_2) = 1$. Plugging these probabilities back into (4) gives us

$$\tilde{P}_\sigma(\tau_\sigma < \infty | \tilde{\lambda}_1, \tilde{\lambda}_2) \geq \epsilon \tilde{P}_\sigma(T(\alpha) < \infty | \tilde{\lambda}_1, \tilde{\lambda}_2) + \tilde{P}_\sigma(T(\alpha) = \infty; \tilde{\lambda}_1, \tilde{\lambda}_2) > \epsilon.$$

However, if for any arbitrary $\tilde{\lambda}_1, \tilde{\lambda}_2$, the probability that $\tau_\sigma < \infty$ is at least ϵ , then it must be that $\tilde{P}_\sigma(\tau_\sigma < \infty) = 1$, a contradiction. $\square$

We now relate the previous lemma to the consumer's beliefs about agent behavior. We show that as agent reputations disappear, the consumer believes with high probability that agents will shirk forever.

Corollary 2. *For all $\epsilon, \alpha > 0$, there exists $\lambda_1, \lambda_2 \in [0, \alpha]^2$ such that $P_\sigma(\tau_\sigma = \infty | \lambda_1, \lambda_2) \geq 1 - \epsilon$.*

Proof. Let $\zeta \leq \frac{1-\epsilon}{(1-\alpha)^2} - 1$. By Lemma 6, there exists $\lambda_1, \lambda_2 \in [0, \alpha]^2$ such that $\tilde{P}_\sigma(\tau_\sigma = \infty | \lambda_1, \lambda_2) \geq 1 - \zeta$. By the law of total probability for any measurable event A , $P_\sigma(A | \lambda_1, \lambda_2) \geq (1 - \lambda_1)(1 - \lambda_2)\tilde{P}_\sigma(A | \lambda_1, \lambda_2)$. Therefore

$$P_\sigma(\tau_\sigma = \infty | (\lambda_1, \lambda_2)) \geq (1 - \lambda_1)(1 - \lambda_2)(1 - \zeta) \geq (1 - \alpha)^2 \frac{1 - \epsilon}{(1 - \alpha)^2} = 1 - \epsilon.$$

□

Corollary 2 proves that as reputations vanish, there must be some pair of reputations for which the consumer believes that strategic agents will shirk forever in the continuation game. Next we consider pivotal reputations: λ_1, λ_2 such that λ_1 is greater than λ_2 , but if output is below y_* , then the consumer's updated belief that agent 1 is reliable will fall below λ_2 . We show that there exists such a reputation pair at which the consumer believes with probability at least $1 - \epsilon$ that $\tau_\sigma = \infty$.

Lemma 7. *For any $\alpha, \gamma, \epsilon > 0$, there exists a reputation pair $(\lambda_1, \lambda_2) \in [0, \alpha]^2$ such that*

$$P_\sigma(\tau_\sigma = \infty | \lambda_1, \lambda_2) > (1 - \epsilon)(1 - \alpha)^2,$$

and

$$\lambda_1 \geq \lambda_2 \geq \frac{\lambda_1 f_1(y)}{\lambda_1 f_1(y) + (1 - \lambda_1) f_0(y)} \quad \text{for all } y \leq y_* - \gamma.$$

Proof. By Lemma 6, for any $\alpha, \epsilon > 0$ there exists a continuation history inducing $(\lambda_1, \lambda_2) \in [0, \alpha]^2$ such that

$$\tilde{P}_\sigma(\tau_\sigma = \infty | \lambda_1, \lambda_2) > 1 - \epsilon.$$

Fix such a pair and consider the continuation game.

The proof proceeds in four steps.

Step 1: Reputations become marginal. Define T as the first time t such that: (i) $\lambda_1^t \geq \lambda_2^t$, (ii) $\lambda_1^t \leq \alpha$, and (iii) the worst possible signal would reverse the ranking, i.e.

$$\lambda_2^t \geq \frac{\lambda_1^t f_1(0)}{\lambda_1^t f_1(0) + (1 - \lambda_1^t) f_0(0)}.$$

We claim that

$$\tilde{P}_\sigma(T < \infty | \tau_\sigma = \infty; \lambda_1, \lambda_2) = 1.$$

Since monitoring is imperfect and types are independent, a player's reputation remains fixed at a strictly positive value in any period in which he is not hired. Hence, if both reputations converge to zero, both agents must be hired infinitely often.

By Lemma 5, if both reputations converge to 0, then $\tau_\sigma = \infty$ almost surely. Then agent i is hired only if his reputation exceeds agent $-i$'s reputation. Thus, if both agents are hired infinitely often, the ranking of reputations must reverse infinitely often almost surely.

By MLRP, a reversal in the ranking of reputations occurs only if the advantaged agent sees a decrease in his reputation via a sufficiently low output realization. Since reputations converge to zero, for any $\alpha > 0$ there exists a time at which such a reversal occurs with $\lambda_1^t \leq \alpha$. To summarize, $\tilde{P}_\sigma(T < \infty | \lambda_1^t, \lambda_2^t \rightarrow 0) = 1$. By Lemma 5, we know that

$$\tilde{P}_\sigma(\lim_{t \rightarrow \infty} \max\{\lambda_1^t, \lambda_2^t\} = 0 \mid \tau_\sigma = \infty) = 1.$$

This then implies that $\tilde{P}_\sigma(T < \infty | \tau_\sigma = \infty) = 1$.

Step 2: The event $T < \infty$ occurs with high probability. We now show that

$$\tilde{P}_\sigma(T < \infty \mid \lambda_1, \lambda_2) > 1 - \epsilon.$$

By the law of total probability,

$$\tilde{P}_\sigma(T < \infty \mid \lambda_1, \lambda_2) \geq \tilde{P}_\sigma(T < \infty \mid \tau_\sigma = \infty; \lambda_1, \lambda_2) \tilde{P}_\sigma(\tau_\sigma = \infty \mid \lambda_1, \lambda_2).$$

The first term equals 1 by Step 1, while the second exceeds $1 - \epsilon$ by construction. Hence the claim follows.

Next, we show that conditional on $T < \infty$, the probability of perpetual shirking remains high. Suppose that for all λ'_1, λ'_2 such that $\alpha \geq \lambda'_1 \geq \lambda'_2 \geq \frac{\lambda'_1 f_1(0)}{\lambda'_1 f_1(0) + (1 - \lambda'_1) f_0(0)}$, $\tilde{P}_\sigma(\tau_\sigma = \infty \mid \lambda'_1, \lambda'_2) < 1 - 2\epsilon$. This implies that

$$\tilde{P}_\sigma(\tau_\sigma = \infty \mid T < \infty; \lambda_1, \lambda_2) < 1 - 2\epsilon.$$

Then

$$\begin{aligned} \tilde{P}_\sigma(\tau_\sigma = \infty \mid \lambda_1, \lambda_2) &= \tilde{P}_\sigma(\tau_\sigma = \infty \mid T < \infty) \tilde{P}_\sigma(T < \infty) \\ &\quad + \tilde{P}_\sigma(\tau_\sigma = \infty \mid T = \infty) \tilde{P}_\sigma(T = \infty) \\ &\leq (1 - 2\epsilon) + \epsilon = 1 - \epsilon, \end{aligned}$$

a contradiction. Therefore, for any $\epsilon > 0$, there must exist a reputation pair λ'_1, λ'_2 such that $\alpha \geq \lambda'_1 \geq \lambda'_2 \geq \frac{\lambda'_1 f_1(0)}{\lambda'_1 f_1(0) + (1 - \lambda'_1) f_0(0)}$ such that

$$\tilde{P}_\sigma(\tau_\sigma = \infty \mid \lambda'_1, \lambda'_2) > 1 - \epsilon.$$

Step 3: Existence of a marginal state with high continuation probability.

We now show that for any $\epsilon, \gamma > 0$, there exists a pair (λ_1, λ_2) such that:

$$1. \alpha \geq \lambda_1 \geq \lambda_2 \geq \frac{\lambda_1 f_1(y)}{\lambda_1 f_1(y) + (1 - \lambda_1) f_0(y)} \text{ for all } y \leq y_* - \gamma \quad (C1)$$

$$2. \tilde{P}_\sigma(\tau_\sigma = \infty \mid \lambda_1, \lambda_2) > 1 - \epsilon \quad (C2)$$

Step 2 demonstrated that there exists λ'_1, λ'_2 such that $\tilde{P}_\sigma(\tau_\sigma = \infty \mid \lambda'_1, \lambda'_2) > 1 - \epsilon$ and $\lambda'_1 \geq \lambda'_2 \geq \frac{\lambda'_1 f_1(0)}{\lambda'_1 f_1(0) + (1 - \lambda'_1) f_0(0)}$. Let p_γ define the minimal probability of transitioning from a state λ'_1, λ'_2 for which $\lambda'_1 \geq \lambda'_2 \geq \frac{\lambda'_1 f_1(0)}{\lambda'_1 f_1(0) + (1 - \lambda'_1) f_0(0)}$ to a state λ_1, λ_2 that satisfies C1.⁹

Suppose for contradiction that for all (λ_1, λ_2) satisfying (C1),

$$\tilde{P}_\sigma(\tau_\sigma = \infty \mid \lambda_1, \lambda_2) \leq 1 - \epsilon/p_\gamma.$$

This implies

$$\tilde{P}_\sigma(\tau_\sigma = \infty \mid \lambda'_1, \lambda'_2) \leq p_\gamma \cdot (1 - \epsilon/p_\gamma) + (1 - p_\gamma) = 1 - \epsilon,$$

contradicting Step 2. Therefore, there exists (λ_1, λ_2) satisfying (C1) and (C2).

Step 4: Conclusion. Consider λ_1, λ_2 satisfying C1 and C2.

Moreover, the continuation probability of perpetual shirking exceeds $1 - \epsilon$. Applying the same argument as in Corollary 3.

$$P_\sigma(\tau_\sigma = \infty \mid \lambda_1, \lambda_2) > (1 - \epsilon)(1 - \alpha)^2,$$

which completes the proof. $\square$

The previous claim tells us that for any $\alpha, \epsilon, \gamma > 0$, there exists a reputation pair λ_1, λ_2 such that three things hold:

1. The consumer believes with probability at least $1 - \epsilon$ that agents will shirk forever: $P_\sigma(\tau_\sigma = \infty \mid \lambda_1, \lambda_2) > 1 - \epsilon$.
2. Reputations are bounded by α : $\lambda_1, \lambda_2 < \alpha$.
3. Reputations are close, i.e. $\lambda_1 \geq \lambda_2$, but for any output realization y below $y_* - \gamma$, $\lambda_2 \geq \frac{\lambda_1 f_1(y)}{\lambda_1 f_1(y) + (1 - \lambda_1) f_0(y)}$.

⁹It is straightforward to show that $p_\gamma > 0$.

We now show that at such a reputation pair, agent 1 has a strictly profitable deviation. The first step is to show that agent 1's beliefs about future behavior do not differ too far from those of the consumer.

Let $\tilde{P}_\sigma^1(\cdot|\lambda_1, \lambda_2)$ denote the probability kernel over outcomes conditional on agent 1 being strategic and the current reputation state being λ_1, λ_2 , and let $\hat{P}_\sigma^1$ denote the distribution conditional on agent 1 being reliable. $\tilde{P}_\sigma^1$ represents agent 1's beliefs about given that he is strategic. Importantly, for any measurable event X , $P_\sigma(X|\lambda_1, \lambda_2) = \lambda_1 \hat{P}_\sigma^1(X|\lambda_1, \lambda_2) + (1 - \lambda_1) \tilde{P}_\sigma^1(X|\lambda_1, \lambda_2)$.

Lemma 8. *For all $\epsilon > 0$, there exists $\lambda_1, \lambda_2 \in [0, \epsilon/2]^2$, and $\tilde{P}_\sigma^1(\tau_\sigma = \infty|\lambda_1, \lambda_2) \geq 1 - \epsilon$.*

Proof. We know for any $\epsilon > 0$, there exists λ_1, λ_2 such that $P_\sigma(\tau_\sigma = \infty|\lambda_1, \lambda_2) > 1 - \epsilon/2$ and $\lambda_1 < \epsilon/2$. This implies

$$\begin{aligned} \tilde{P}_\sigma^1(\tau_\sigma = \infty|\lambda_1, \lambda_2) &= \frac{P_\sigma(\tau_\sigma = \infty|\lambda_1, \lambda_2) - \lambda_1 \hat{P}_\sigma^1(\tau_\sigma = \infty|\lambda_1, \lambda_2)}{1 - \lambda_1} \\ &\geq P_\sigma(\tau_\sigma = \infty|\lambda_1, \lambda_2) - \frac{\epsilon}{2} \hat{P}_\sigma^1(\tau_\sigma = \infty|\lambda_1, \lambda_2) \\ &> 1 - \epsilon/2 - \epsilon/2 = 1 - \epsilon \end{aligned}$$

□

We have now shown that for any $\epsilon, \gamma > 0$ there exists a reputation pair (λ_1, λ_2) at which three things hold:

1. λ_1 and λ_2 are less than $\epsilon/2$.
2. Agent 1 believes with probability at least $1 - \epsilon$ that $\tau_\sigma = \infty$.
3. $\lambda_1 \geq \lambda_2$ but for any output below $y_* - \gamma$, $\lambda_2 \geq \frac{\lambda_1 f_1(y)}{\lambda_1 f_1(y) + (1 - \lambda_1) f_0(y)}$.

Point 2 states that when reputations are equal to (λ_1, λ_2) , agent 1 believes that agents will always shirk forever with high probability, meaning he will only be hired if his reputation exceeds agent 2's. We complete the proof of theorem 2 by showing that at (λ_1, λ_2) , agent 1 has a profitable deviation.

Since agent 1 believes with high probability that he will only be hired if his reputation exceeds agent 2's reputation, it is convenient to summarize his payoffs in terms of the ratio of posterior odds. Define the function $z : [0, 1]^2 \rightarrow [0, \infty)$, which is equal to the ratio of likelihood ratios between the two agent reputations

$$z(\lambda_1, \lambda_2) = \frac{\frac{\lambda_1}{1 - \lambda_1}}{\frac{\lambda_2}{1 - \lambda_2}}$$

Consider the continuation game starting at λ_1, λ_2 . We let $z_t = z(\lambda_1^t, \lambda_2^t)$. Agent 1 assigns probability at least $1 - \epsilon$ to the event that, in every future period, agent 2 shirks and agent 1 is hired only if $z_t \geq 1$. Conditional on this belief, agent 1's payoff equals

$$(1 - \delta) \sum_t \delta^t \mathbb{1}\{z_t \geq 1\}$$

We assume ties go in favor of agent 1, which is without loss since ties occur with probability 0.

We next explain the transition dynamics of z . Suppose $z \geq 1$. The consumer will hire agent 1. Then agent 2's reputation will stay static, whereas agent 1's reputation will be updated via Bayes' rule. Then if output y is realized, z will take new value $g_1(z, y) = z \frac{f_1(y)}{f_0(y)}$. By similar logic, $z < 1$ implies agent 2 is hired, so after output y , z will take on new value $g_2(z, y) = z \frac{f_0(y)}{f_1(y)}$. We summarize by defining the transition function g as:

$$g(z, y) = \begin{cases} g_1(z, y) & z \geq 1 \\ g_2(z, y) & z < 1 \end{cases}$$

To analyze incentives, we define agent 1's payoff if agent 2 is strategic and agents shirk forever. Suppose agent 2 is strategic, and $\tau_\sigma = \infty$. We can express agent 1's payoff in this event recursively. Let $v : [0, \infty) \rightarrow [0, 1]$ denote agent 1's payoff when the relative likelihood ratio is z , agents are strategic, and they shirk forever. If $z \geq 1$, then he obtains payoff $1 - \delta$ today, and since he shirks, his continuation value is $E_0[v(g_1(z, y))]$. His total payoff is then $1 - \delta + \delta E_0[v(g_1(z, y))]$. If $z < 1$, then agent 2 is hired. Since agent 2 shirks, agent 1's expected future payoff equals $E_0[v(g_2(z, y))]$. Therefore, v is given by:

$$v(z) := (\mathbb{1}\{z \geq 1\})(1 - \delta + \delta E_0[v(g_1(z, y))]) + (\mathbb{1}\{z < 1\})(\delta E_0[v(g_2(z, y))])$$

We show via the contraction mapping theorem that there exists unique v satisfying the above recursive definition. In addition, v satisfies the following 3 properties.

1. **(P1) Symmetry:** $\forall z > 0, z \neq 1, v(z) = 1 - v(\frac{1}{z})$.
2. **(P2) Monotonicity:** v is increasing in z .
3. **(P3) Better to be ahead:** There exists $\underline{\delta} < 1$ such that for all $\delta \geq \underline{\delta}, z \geq 1 \Rightarrow v(z) \geq 1/2 + \frac{1-\delta}{2}$.

We reserve the proof for the end of the appendix. Importantly, we assume for the remainder of the proof that $\delta \geq \underline{\delta}$.

Note that if $v(z) \geq 1/2 + \frac{1-\delta}{2}$ for $z \geq 1$ and v is symmetric, then for all $z' < 1, v(z') \leq 1/2 - \frac{1-\delta}{2}$,

therefore $v(z) - v(z') \geq 1 - \delta$. This particularly implies that for $z \geq 1$, the expected continuation payoff $E_0[v(g_1(z, y))]$ is greater than or equal to the expected continuation payoff $E_0[v(g_2(z', y))]$ for $z' < 1$.

v represents agent 1's payoff conditional on agent 2 being strategic and $\tau_\sigma = \infty$. At reputation pair λ_1, λ_2 , agent 2 is strategic with probability $1 - \lambda_2$, and agent 1 assigns probability at least $1 - \epsilon$ to $\tau_\sigma = \infty$. Therefore, agent 1's expected payoff at λ_1, λ_2 equals

$$(1 - x) [(1 - \lambda_2) (1 - \delta + \delta E_0[v(g_1(z, y))]) + \lambda_2 \mu] + xr$$

where x is less than or equal to ϵ , $z(\lambda_1, \lambda_2) \geq 1$, and μ, r are values between 0 and 1. Compare this to agent 1's payoff if he deviates to high effort:

$$(1 - x) [(1 - \lambda_2) ((1 - \delta)(1 - c) + \delta E_1[v(g_1(z, y))]) + \lambda_2 \mu'] + xr'$$

Subtracting his payoff under low effort from this gives us that deviating is profitable if and only if

$$(1 - x)(1 - \lambda_2) [(1 - \delta)(-c) + \delta E_1[v(g(z, y))] - \delta E_0[v(g(z, y))]] \geq \lambda_2(\mu - \mu') + x(r - r') \quad (5)$$

Focus attention on the difference in expected continuation payoffs: $E_1[v(g(z, y))] - E_0[v(g(z, y))]$. By Lemma 7, there exists $\hat{y} \in [y_* - \gamma, y_*)$ such that $g(z, y) \geq 1$ if and only if $y \geq \hat{y}$. Under effort e , $g(z, y) < 1$ with probability $F_e(\hat{y})$. With complementary probability $g(z, y) \geq 1$. Thus:

$$E_e[v(g(z, y))] = (1 - F_e(\hat{y}))A + F_e(\hat{y})B$$

where

$$\begin{aligned} A &:= 1 - \delta + \delta E_0[v(g(z, y), y') | y \geq \hat{y}], \text{ and} \\ B &:= \delta E_0[v(g(z, y), y') | y < \hat{y}]. \end{aligned}$$

A is the expected future payoff in the event that current period output y exceeds $\hat{y}$, whereas B is the same object conditioned on output y being less than $\hat{y}$. By property (P3) of the value function v , $A - B \geq 1 - \delta$.

Taking the difference $E_1[v(g(z, y))] - E_0[v(g(z, y))]$ therefore yields

$$(F_0(\hat{y}) - F_1(\hat{y}))[A - B] \geq (F_0(\hat{y}) - F_1(\hat{y}))(1 - \delta).$$

The last step to note is that if $F_0(y_*) - F_1(y_*) > c$, then there exists $\underline{\delta}'$ such that if $\delta \geq \underline{\delta}'$, then $F_0(y_*) - F_1(y_*) > \frac{c}{\delta}$. Let $\delta \geq \max\{\underline{\delta}, \underline{\delta}'\}$. Additionally, for γ small enough,

$$E_1[v(g(z, y))] - E_0[v(g(z, y))] \geq (F_0(\hat{y}) - F_1(\hat{y}))(1 - \delta) > \frac{(1 - \delta)c}{\delta}.$$

Plugging this into the bracketed expression on the left-side of equation (5) gives us that

$$(1 - \delta)(-c) + \delta E_1[v(g(z, y))] - \delta E_0[v(g(z, y))] = (1 - \delta)(-c) + \delta \frac{(1 - \delta)c}{\delta} = 0.$$

This means that the left-side of (5) is greater than or equal to $(1 - x)(1 - \lambda_2)\beta$ where $\beta > 0$. To prove whether there is a profitable deviation to high effort, it is sufficient to prove that

$$(1 - \lambda_2)(1 - x)\beta \geq \lambda_2(\mu - \mu') + x(r - r')$$

Since $\mu, \mu', r, r' \in [0, 1]$, and $x, \lambda_2 < \epsilon$, we can choose ϵ small enough such that the above inequality holds. This proves that a deviation to high effort is strictly profitable, which completes the proof of Theorem 2.

B.2.1 Properties of value function v

B.2.2 Uniqueness

The function v can be recursively defined as:

$$v(z) = \mathbb{1}\{z \geq 1\}(1 - \delta + \delta E_0[v(g(z, y))]) + \mathbb{1}\{z < 1\}(\delta E_0[v(g(z, y))]).$$

We show that there is a unique v that meets the above relation. Let $\mathcal{F}$ be the set of functions from $[0, \infty)$ to $[0, 1]$.

Define the norm $\|\cdot\|$ on $\mathcal{F}$ as:

$$\|w\| = \sup_{z \in [0, \infty)} |w(z)|$$

Such a norm is well defined because the range of each function in $\mathcal{F}$ is bounded by 0 and 1. Give X the distance induced by the sup norm. The space $\mathcal{F}$ is a complete metric space (See Stokey,

Lucas Jr, and Prescott (1989)).

On $\mathcal{F}$, we define the Bellman operator Γ . For $w \in \mathcal{F}$, and $z \in [0, \infty)$, let $\Gamma(w)(z) =$

$$\mathbb{1}\{z \geq 1\} (1 - \delta + \delta E_0[w(g(z, y))]) + \mathbb{1}\{z < 1\} (\delta E_0[w(g(z, y))])$$

we show that Γ is a self map and a contraction.

Self map

$\forall w \in \mathcal{F}$, that $\Gamma(w)$ outputs values in $[0, 1]$ is obvious. All that is left to show is that Γ is a contraction.

Contraction Let $w, w' \in \mathcal{F}$. We show that $\|\Gamma w - \Gamma w'\| \leq \delta \|w - w'\|$. Suppose first that $z \geq 1$.

$$\begin{aligned} |\Gamma w(z) - \Gamma w'(z)| &= \delta |E_0 [w(g(z, y)) - w'(g(z, y))]| \\ &\leq \delta E_0 [|w(g(z, y)) - w'(g(z, y))|] \\ &\leq \delta E_0 [|w - w'|] = \delta \|w - w'\| \end{aligned}$$

The same argument applies for $z < 1$. Therefore Γ is a contraction. By the contraction mapping theorem, Γ has a unique fixed point v .

B.2.3 Properties of v

We show 2 important properties of v . The approach we use is to consider a function $w \in \mathcal{F}$ satisfying each property and then show the property is preserved under Γ .

1. **Symmetry:** $\forall z \neq 0, 1, v(z) = 1 - v(\frac{1}{z})$

To show this, we suppose that a function $w \in \mathcal{F}$ satisfies symmetry. Let $z > 1$. $\Gamma w(z) + \Gamma w(1/z) = 1 - \delta + \delta E_0[w(z \frac{f_1(y)}{f_0(y)})] + \delta E_0[w(\frac{f_0(y)}{z f_1(y)})] = 1 - \delta + \delta E_0[w(z \frac{f_1(y)}{f_0(y)}) + w(\frac{f_0(y)}{z f_1(y)})]$. Since the output distribution is atomless, $w(z \frac{f_1(y)}{f_0(y)}) + w(\frac{f_0(y)}{z f_1(y)}) = 1$ almost surely. Therefore,

$$1 - \delta + \delta E_0[w(z \frac{f_1(y)}{f_0(y)}) + w(\frac{f_0(y)}{z f_1(y)})] = 1 - \delta + \delta = 1.$$

2. **Monotonicity:** Suppose w is monotone and symmetric. We show monotonicity is preserved under Γ . Let $z > z' \geq 1$.

$$\Gamma w(z) = 1 - \delta + \delta E_0[w(g(z, y))] \geq 1 - \delta + \delta E_0[w(g(z', y))] = \Gamma w(z').$$

The inequality follows by monotonicity of w . The same argument follows for $z < z' < 1$. In the case of $z > 1 > z'$, we will show that $v(z) \geq 1/2 + (1-\delta)/2$. By symmetry of v , this would imply $v(z') \leq 1/2 + \frac{1-\delta}{2}$, which would prove global monotonicity. To show this, consider a function w such that for all $z \geq 1$, $w(z) \geq 1/2 + \beta$ for $\beta > \frac{1-\delta}{2}$.

Assume furthermore that for any $z < 1$ such that $p(z) := \int_{y: g_2(z,y) \geq 1} f_0(y) dy > 0$, i.e. there is positive probability that bad output will lead to z transitioning to a value above 1, $w(z) \geq \delta p(z)(1/2 + \beta)$. We first argue that the same is true for $\Gamma w(z)$. To see this, note that:

$$\begin{aligned} \Gamma w(z) &= \delta \left(\int_{y: g(z,y) \geq 1} w(g(z,y)) f_0(y) dy + \int_{y: g(z,y) < 1} w(g(z,y)) f_0(y) dy \right) \\ &\geq \delta \int_{y: g(z,y) \geq 1} w(g(z,y)) f_0(y) dy \\ &\geq \delta \int_{y: g(z,y) \geq 1} (1/2 + \beta) f_0(y) dy \\ &= \delta p(z)(1/2 + \beta). \end{aligned}$$

Since $\delta < 1$, there exists $k(z)$ such that $p(z) \geq \delta^{k(z)}$. This value $k(z)$ is decreasing in z by monotonicity of w . It follows, then, that there exists $k^* < \infty$ such that for any $z \geq 1$,

$$\mathbb{E}_0[w(g(z,y)) | g(z,y) < 1] \geq \delta^{k^*} (1/2 + \beta).$$

Next, consider $\Gamma w(z)$ for $z \geq 1$.

$$\Gamma w(z) \geq 1 - \delta + \delta P_0(g(z,y) \geq 1)(1/2 + \beta) + \delta^{k^*} P_0(g(z,y) < 1)(1/2 + \beta).$$

We argue that

$$\Gamma w(z) \geq 1 - \delta + \delta P_0(g(z,y) \geq 1)(1/2 + \beta) + \delta^{k^*} (1/2 + \beta) \geq 1/2 + \sqrt{\frac{\beta(1-\delta)}{2}}.$$

Rearranging, we get that this is true if and only if

$$P_0(g(z, y) \geq 1) + \delta^{k^*} P_0(g(z, y) < 1) \geq \frac{\delta - 1/2 + \sqrt{\frac{\beta(1-\delta)}{2}}}{1/2 + \beta}. \quad (6)$$

We focus attention on the right-side of (6):

$$\frac{\delta - 1/2 + \sqrt{\frac{\beta(1-\delta)}{2}}}{1/2 + \beta} \leq \frac{\delta - 1/2 + \sqrt{\frac{\beta(1-\delta)}{2}}}{1/2} = 2\delta - 1 + 2\sqrt{\frac{\beta(1-\delta)}{2}} \leq 2\delta - 1 + 2\sqrt{\beta(1-\delta)},$$

where the first inequality follows from $\beta \geq 0$. Next, notice that for any $\epsilon > 0$, we can choose $\frac{1-\delta}{2} < \beta < 1 - \delta$ such that $\sqrt{\beta(1-\delta)} \leq 1 - \delta - \epsilon$, which implies

$$\frac{\delta - 1/2 + \sqrt{\frac{\beta(1-\delta)}{2}}}{1/2 + \beta} \leq 2\delta - 1 + 2\sqrt{\beta(1-\delta)} \leq 1 - 2\epsilon.$$

Now for the other side (6), notice that $P_0(g(z, y) \geq 1) + \delta^k P_0(g(z, y) < 1) \geq (1 - F_0(y_*)) + \delta^k F_0(y_*)$. This is because the probability that $g_1(y, z) \geq 1$ is greater than or equal to the probability that output exceeds y_* , the threshold for one's reputation to improve. Notice, also, then that for any $\epsilon > 0$, there exists $\underline{\delta} < 1$ such that for all $\delta \geq \underline{\delta}$,

$$1 - F_0(y_*) + \delta^k F_0(y_*) \geq 1 - 2\epsilon \geq 2\delta - 1 + 2\sqrt{\beta(1-\delta)}, \text{ which implies that}$$

$$\Gamma w(z) \geq 1/2 + \sqrt{\frac{\beta(1-\delta)}{2}}.$$

Notice that since $\beta > \frac{1-\delta}{2}$ by assumption, we have that for all $z \geq 1$, $\Gamma w(z) \geq 1/2 + \frac{1-\delta}{2}$. By the symmetry of Γw , this implies that for all $z' < 1$, $\Gamma w(z') \leq 1/2 - \frac{1-\delta}{2}$, which therefore implies that Γw is monotone.

Next, we construct the recursive sequence:

$$\beta_0 = \beta \in (\frac{1-\delta}{2}, 1 - \delta), \text{ and } \forall n \geq 1, \beta_n = \sqrt{\frac{\beta_{n-1}(1-\delta)}{2}}.$$

It can be easily shown that $(\beta_n)_{n \in \mathbb{N}}$ is a decreasing sequence that is bounded below by its limit: $\frac{1-\delta}{2}$. For notational purposes, note for any function $w \in \mathcal{F}$, we define $\Gamma^n w$ to be the operator Γ applied to w n times.

We now make the following claim:

Claim 7. Consider a monotone, symmetric function $w : [0, \infty) \rightarrow [0, 1]$ such that $w(z) \geq 1/2 + \beta_0$ for all $z \geq 1$, and $w(z) \geq \delta^{k(z)}(1/2 + \beta_0)$ for all $z < 1$ such that there is probability under low effort $p(z) > 0$ that $g_2(z, y) \geq 1$. Then for all $\delta \geq \underline{\delta}$ and $n \in \mathbb{N}$, $\Gamma^n w(z) \geq 1/2 + \beta_n$ for all $z \geq 1$ and $\Gamma^n w(z) \geq \delta^{k(z)}(1/2 + \beta_n)$ for all $z < 1$ such that there is positive probability under low effort that $g(z, y) \geq 1$.

Proof. The claim is trivially true for $n = 0$. The argument above the claim showed that it is also true for $n = 1$. The proof for all remaining n can be done via induction. Suppose the claim is true for $n \in \mathbb{N}$. Consider $z < 1$ such that

$$\int_{y: g_2(z, y) \geq 1} f_0(y) dy > 0.$$

It follows that

$$\begin{aligned} \Gamma^{n+1} w(z) &= \delta \int_Y \Gamma^n w(g(z, y)) f_0(y) dy \\ &\geq \delta \int_{y: g(z, y) \geq 1} \Gamma^n w(g(z, y)) f_0(y) dy \\ &\geq \delta \int_{y: g(z, y) \geq 1} (1/2 + \beta_n) f_0(y) dy \\ &= \delta p(z) (1/2 + \beta_n) \\ &\geq \delta^{k(z)} (1/2 + \beta_{n+1}). \end{aligned}$$

Therefore, for all $z \geq 1$,

$$\Gamma^{n+1} w(z) \geq 1 - \delta + \delta P_0(g(z, y) \geq 1) (1/2 + \beta_n) + \delta^{k^*} P_0(g(z, y) < 1) (1/2 + \beta_n)$$

Following similar steps as for the β_0 step, we get that $\Gamma^{n+1} w(z) \geq 1/2 + \beta_{n+1}$ if

$$1 - F_0(y_*) + \delta^{k^*} F_0(y_*) \geq 2\delta - 1 + 2\sqrt{\beta_n(1 - \delta)}.$$

Since $\beta_n < \beta_0$, we know that $2\delta - 1 + 2\sqrt{\beta_n(1 - \delta)} < 2\delta - 1 + 2\sqrt{\beta_0(1 - \delta)} < 1 - 2\epsilon$. Then

$\delta \geq \underline{\delta}$ implies $1 - F_0(y_*) + \delta^{k^*} F_0(y_*) \geq 2\delta - 1 + 2\sqrt{\beta_n(1-\delta)}$. This completes the proof. $\square$

Assume $\delta \geq \underline{\delta}$. For any monotone symmetric function w that satisfies the properties listed in claim 7, $\Gamma^n w(z) \geq 1/2 + \beta_n \geq 1/2 + \frac{1-\delta}{2}$ for all $n \in \mathbb{N}$ and $z \geq 1$. Therefore, taking the limit as $n \rightarrow \infty$, $v(z) \geq 1/2 + \frac{1-\delta}{2}$ for any $z \geq 1$. Furthermore, since v is symmetric, we know that for all $z < 1$, $v(z) \leq 1/2 - \frac{1-\delta}{2}$. This implies that for any $z > 1 > z'$, $v(z) \geq v(z')$, which completes the proof of monotonicity of v .